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Original Article

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A comprehensive analysis of capacity utilization rates of fast-charging stations in shopping malls: advancing sustainable management through real-time capacity analytics

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Abstract

The widespread adoption of electric vehicles has intensified the demand for effective charging infrastructure, particularly in urban shopping malls. As of June 2025, the total number of charging stations installed in Turkey reached 32 920, corresponding to a cumulative installed capacity of 2.38 GW (gigawatts). This paper analyzes key performance metrics such as utilization frequency and capacity efficiency using a unique, monthly socket-level dataset from a designated mall. A key finding is that the maximum utilization rate has remained consistently low at 12%, indicating significant underuse. The results offer strategic insights that inform the optimization of infrastructure, managing energy resources, and planning future investments to improve fast-charging station performance and support the transition to sustainable mobility.

Keywords: charging time and load balancing analysis; electric vehicle user behaviour modelling; fast-charging stations in shopping malls; operational efficiency in charging stations; urban mobility

1 Introduction

Global electric vehicle (EV) sales exceeded 17 million units, capturing over 20% of the automotive market in 2024, with the annual growth alone surpassing the total number of EVs sold worldwide in 2020. The adequacy of public charging infrastructure is closely linked to its charging capacity; notably, the number of ultrafast chargers ($\geq$150 kW) increased by approximately 50% in 2024, now representing nearly 10% of the global public fast-charging network [1]. The recent global proliferation of EVs can be attributed to a series of support policies introduced by China in 2009 [2]. Despite the uninterrupted technological advancement and pervasive adoption of EVs in the contemporary era, it is observed that a subset of users harbours reservations pertaining to the acceptance of EVs, which can be attributed to a multitude of factors [3]. Indeed, a recent survey of EV users conducted by McKinsey revealed that battery capacity, range, costs, and charging infrastructure are among the primary concerns among this demographic [4]. The lack of clarity surrounding the occurrence of fires in EVs, the role of the second-hand market, and the processes involved in breakdown, repair, and maintenance also influence user preferences [5].

As indicated in the *Electric Vehicle and Charging Infrastructure Projection Report* prepared by Energy Market Regulatory Authority (EMRA) in Turkiye, the number of EVs in use is projected to range from 1 779 488 to 4 214 273 by 2035, and in order to accommodate the anticipated number of EVs on the road, it is estimated that 146 916–347 934 charging stations will be required [6]. A further report forecasts that there will be 10 million EVs in use in Turkiye by 2040, with an estimated requirement for approximately 10 million charging stations to facilitate their operation [7, 8]. It is anticipated that 60% of these stations will be situated in public spaces. In the medium and long term, as the use of EVs increases, it is crucial to optimize charging hours and balance electricity consumption in homes, workplaces, and public spaces to minimize the peak load increase in electricity demand. This necessitates a strategic approach to energy management and infrastructure planning.

This study examines the experience with charging stations, which represent the most crucial component of the development of the EV ecosystem, and their impact on user motivations. In this context, potential improvements to the efficiency of existing charging stations and charging processes have been evaluated, beginning with the premise that the location of charging stations, the quality of service, and the cost of charging units are the primary motivators for the use of EVs.

Despite the growing deployment of EV infrastructure, empirical studies that scrutinize temporal usage dynamics within context-specific commercial environments remain virtually absent in Turkiye. This study steps into that void by offering a rare, data-grounded perspective derived from an EV charging station located in a shopping mall—a site of high commercial activity but underexplored in academic discourse. Through the analysis of granular time series data spanning 1 year, the study investigates operational patterns via

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indicators such as usage intensity, average charging duration, capacity occupancy, and concurrency ratios. Rather than merely reporting descriptive outcomes, the work critically interprets these metrics to expose demand asymmetries and latent behavioural tendencies. From this basis, a responsive configuration model is proposed—one that does not assume uniform usage but instead adapts to spatial and temporal variability. In parallel, the study reframes how unit charging cost is conceptualized, arguing for a model that balances user access with operator sustainability. The result is a contribution that not only bridges a documented gap but also challenges assumptions in the infrastructural planning of urban electromobility.

In this paper, a novel dynamic pricing framework is proposed for EV charging infrastructure, with a focus on utilization-based tariff modulation to enhance operational efficiency and consumer advantage. While current regulations in Turkiye permit differential pricing based on power output levels, these mechanisms do not adequately incorporate real-time station availability into pricing strategies. A demand-sensitive pricing algorithm, responsive to temporal station idleness, could enable users to access more economical rates and incentivize optimal infrastructure use. Implementation of such a system requires the integration of live occupancy data via mobile applications; however, reliance on operator-specific platforms presents limitations in interoperability and accessibility. To address this, the study recommends the development of a centralized digital platform aggregating all public charging stations, thereby enabling real-time transparency, fostering price-driven competition, and improving user experience through informed decision-making.

2 A comprehensive review of key findings and emerging trends in the literature

As the penetration of EVs in the global market increases, the planning of charging infrastructures with high accessibility and utilization efficiency designed to meet the energy requirements of these vehicles has become a critical engineering problem. The global EV fleet consumed approximately 180 TWh of electricity in 2024, reflecting an increase of nearly 60% compared to the previous year [9]. Shopping malls constitute strategic venues for the expansion of EV charging infrastructure, particularly for semifast and fast-charging stations. Their suitability is underpinned by two main factors: the relatively long average parking durations, which align well with charging times, and the high user density, which ensures consistent demand and operational efficiency. Moreover, the architectural and spatial characteristics of shopping malls typically provide the necessary infrastructure to support such installations without significant modification.

The role of EV charging strategies extends beyond logistical convenience, positioning them as critical enablers of carbon emission reduction. When aligned with renewable energy sources, smart charging systems significantly lower the carbon intensity of electricity use and support the stability of energy grids. Technologies such as time-responsive algorithms and vehicle-to-grid (V2G) systems allow for interactive energy management, where EVs contribute to both storage and supply functions. In this regard, the design of charging infrastructure is not merely a technical necessity but a strategic environmental instrument. However, the effectiveness of these systems depends on a nuanced understanding of infrastructural capacity and operational dynamics. Moreover, as demonstrated by Wang *et al.* [10], apart from widely adopted measures such as free public charging and purchase tax exemptions, additional financial incentives exert only a marginal influence on consumer willingness to adopt EVs.

As the use of EVs becomes more prevalent and the number of charging stations expands, a greater proportion of drivers will be able to utilize EVs over longer distances. Accordingly, the number of charging units, the utilization rate of charging stations, and the number of EVs in the market are critical for achieving this parity, which serves as an important indicator for the investments to be made for charging infrastructure. Over time, this will allow the capacity utilization rates of charging stations to increase. An increase in the capacity utilization rate is contingent upon a growth in the number of charging stations that outpaces the expansion of the vehicle population within the area where they are deployed [11].

Upon examination of the data presented in the literature, it becomes evident that the capacity utilization rate of charging stations, the majority of which are situated in public areas, falls within the range of 5% to 15% [12]. In the research report prepared by the Shura Energy Transformation Center to determine the utilization rates of charging station infrastructures based on Istanbul in 2021, the return period of a 50-unit charging station investment was calculated as 1.13 years according to the approach where the EV charging station capacity utilization rate was accepted as 12.5%. This was determined using direct current (DC) (>100 kW) fast chargers with two socket outlets [11].

A significant factor to be considered in the identification of charging stations and the optimization of installation approaches is the average parking behaviour of EV users. It is of paramount importance to ascertain the capacity utilization rates of the charging stations to be installed, and this can be achieved by establishing the aforementioned habits. In accordance with the aforementioned, while the average parking times are relatively similar in homes and workplaces in general, the parking times stated as 1–3 hours in shopping malls, 30 minutes to 3 hours in open or closed parking lots, and 30 minutes in intercity rest areas necessitate the implementation of disparate charging levels and infrastructure installations for charging stations [13]. The issue of energy demand in this context is dependent on the habits of EV users and is seen as a potential contributor to the already increasing energy demand in the region. This is particularly relevant in relation to the need for charging stations to be connected to the grid from electricity distribution lines at numerous points.

It is anticipated that the number of charging stations installed for EVs will reach approximately 210 million worldwide by 2035, with the majority of these stations located in major markets such as the EU, the US, and China. China is a global leader in the EV industry, with a significant presence in the infrastructure sector. The country accounts for over 85% of the world's fast chargers and nearly 60% of the world's slow chargers [9]. Moreover, by 2023, the number of public charging stations for EVs (including both alternating current [AC] and DC charging stations) across Europe, including Turkiye, is estimated to be approximately 755 800 [14]. Moreover, the Alternative Fuel Infrastructure Regulation mandates the installation of DC fast-charging infrastructure (with a minimum power output of 150 kW) at a rate of at

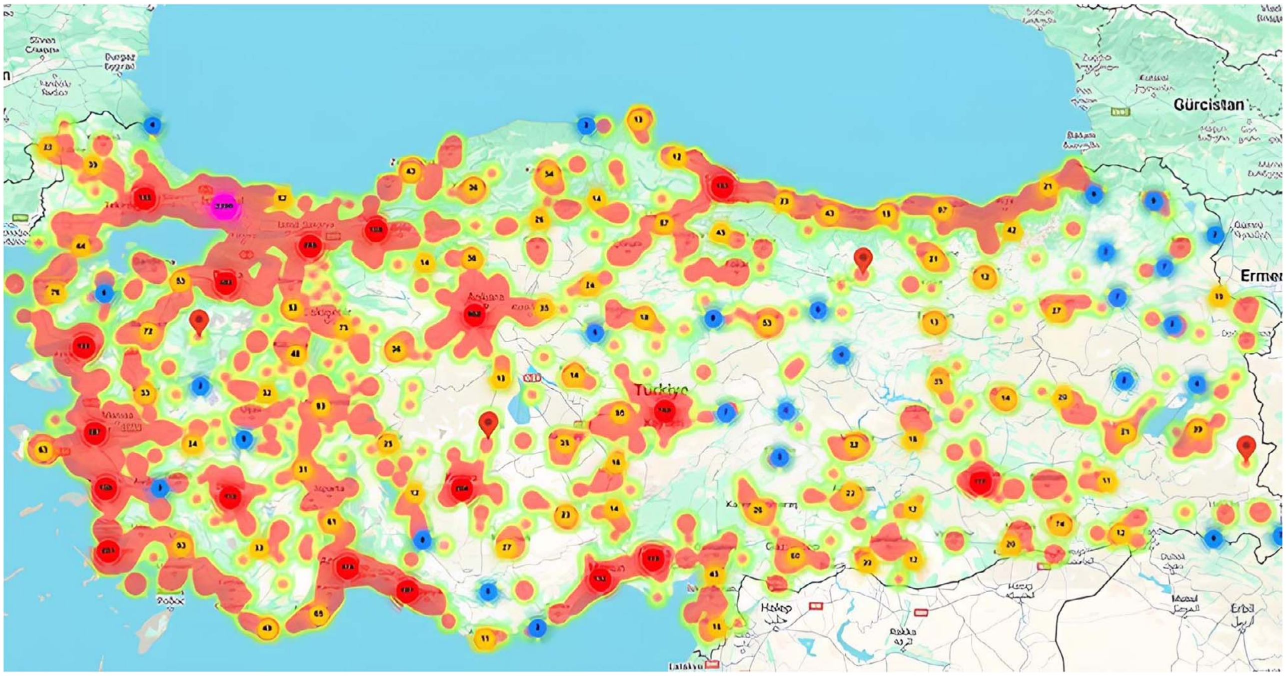

Figure 1. Density map of EV charging station infrastructure in Turkiye. [Source: EVCify]

least one facility every 60 km along the EU Trans-European Transport Network, beginning in 2025. This requirement is anticipated to undergo an increase in the future [15].

The number of electric cars registered in Turkiye during the first 3 months of 2025 was 268 005, of which 1.3% were electric cars [16]. It is anticipated that this value will reach 2 million by 2030, according to a medium-scale scenario model [6].

As illustrated in Fig. 1, the EV charging station infrastructure density map for Turkiye reveals that charging stations are distributed unevenly across the country. These illustrated data are produced by EVCify based on EMRA data and utilize REST Web Services for the inquiry of charging stations in the charging station list.

The EMRA has published its monthly statistics report on the Charging Service Market for June 2025. According to the report, Turkiye's total installed charging station capacity reached 2.38 GW in terms of total socket output power, with 32920 charging stations in operation. By the end of March, the number of AC charging points had reached 19036, while the number of DC charging points had reached 13884 [17].

The total number of charging station licensees in the Turkish charging station market reached 178 at the end of 2024, which is an increase from the previous year. It is estimated that approximately 80% of public charging stations are managed by the current Top 5 providers of the charging network on the market [6]. For new market entrants, the ongoing evolution of EV charging technology, coupled with rising maintenance and rental costs, will constrain profit margins, complicating the process of making investment decisions. In order to address this fundamental issue, it is necessary to develop new business models in conjunction with the provision of a seamless charging experience.

The regulation of service fees for EV users is a complex issue that varies across countries. Pricing models include time-based tariffs, which are charged based on the time spent charging, as well as methods that combine time and unit energy cost at varying rates. Additionally, there are various types of additional charges, such as start fees and charges based on charging time. In Turkiye the charging service is applied in terms of the unit price of energy transferred to the EV, and no additional fee is charged under any other name than the fee calculated on the basis of the charging service price [18]. It should be noted that supplementary fees may be incurred should the socket link not be terminated at the conclusion of the designated charging interval. In Turkiye, different price tariffs are applied for charging stations, with the EMRA holding the authority to set a ceiling price. Despite the assertion that the extant regulations do not encompass a multitime pricing methodology for the charging service fee that would align with the actual costs of the energy network and the market, it can be posited that this does not present an insurmountable challenge.

A study conducted in Norway indicates that in this country, where the EV ecosystem is mature, EV owners tend to charge their vehicles during peak hours, particularly around 18:00, irrespective of the time-varying effects of electricity tariffs or incentives [19]. In this context, Hildermeier *et al.* proposed the concept of smart charging. The concept is generally defined as a means of shifting charging times without compromising the needs of EV owners during periods when electricity generation and transmission costs are low. Additionally, it is a means of reducing the operating costs of EVs and making the charging process profitable for users and the electricity grid by shifting energy demand on the grid during peak hours of energy consumption [20].

The integration of EV charging stations into the electrical grid is regarded as a substantial challenge. The autonomy of EV charging regimes with regard to the requirements of the electricity distribution grid can have adverse effects on the operation of the latter. In order to mitigate the aforementioned impacts and effectively manage the supplementary energy demand from EVs, it is crucial to implement intelligent charging strategies and business models. This will facilitate the provision of cost-effective charging solutions for EV users and enhance the efficiency of grid utilization.

Shura's report 'Transforming Turkiye's transportation sector: Impacts of EVs on the Turkish distribution grid', published by Shura, analysed how EVs can be integrated into distribution grids in Turkey and how different charging optimization strategies can help mitigate potential problems. It was posited that the integration of EV's into the grid may present certain challenges, which could potentially lead to a significant increase in the required investment in grid infrastructure. The primary challenges identified were the absence of a comprehensive plan for the scheduling of charging times, the determination of optimal charging locations, and the estimation of the necessary charging station capacity. The implementation of multitime electricity pricing tariffs, the adoption of innovative business models and technologies in charging station operations, along with the strategic planning of charging points and grid locations will play a crucial role in minimizing stress on the grid while maintaining the comfort and convenience of vehicle owners [7, 8].

A review of EV charging station operations in Europe revealed the existence of five different pricing models. These models were identified through the examination of approximately 3 million charging transactions from diverse countries. The aforementioned categories are detailed as follows: free, fixed fee, time based, energy based, and hybrid. It was indicated that AC/DC charging stations are priced based on a range of performance indicators, including charging and idle times, energy capacity transferred, average power, and revenue generated. It has been posited that time-based pricing models may facilitate greater accessibility to public charging infrastructure, given that the connection time per charging event can be reduced by approximately 50% in comparison to alternative pricing models [21].

This study employs survey data from four Chinese cities with a high prevalence of EVs to quantitatively analyse the factors influencing the satisfaction of EV users with the existing charging infrastructure. The study indicates that individuals with higher income levels, greater tolerance for waiting in queues, and longer driving distances are more likely to select public charging infrastructure for their charging preferences, whereas those who favour night-time charging and are sensitive to charging prices tend to be more satisfied with private charging infrastructure [22].

The implementation of comprehensive and efficiently serviced charging infrastructure networks has the potential to facilitate longer-distance travel for EV drivers, thereby reducing the phenomenon of range anxiety [23]. The results of the survey conducted by Sun *et al.* indicate that 80% of EV users prefer to utilize charging stations situated within public parking facilities [24]. The extant literature indicates that a significant proportion of EV users will opt for private charging infrastructures based on individual use if the requisite conditions are met. This would effectively elevate the home charging option to a prominent position [25].

In a comprehensive literature review on consumer preferences for public and private charging infrastructure, Hardman *et al.* identified four key factors that significantly influence EV users' charging preferences: driving experience, charging time, charging station location, and charging service unit price [26].

The optimal time interval for charging EVs has a favourable and substantial effect on consumer satisfaction. It is observed that vehicle owners who prefer to charge their vehicles at night tend to favour private charging stations over public charging points, particularly in the context of existing charging infrastructures. In consideration of the aforementioned charging timeframe, which currently coincides with the peak hours of electricity consumption, it is asserted that alterations to electricity pricing and the implementation of guidance for EV users are necessary to mitigate the impact on the electricity grid [27].

A substantial body of literature demonstrates that the cost of charging has a significant influence on the satisfaction of EV users [28]. Moreover, Hardman *et al.* posit that the advancement of EV charging infrastructure can be propelled by a multitude of stakeholders, including policy makers, shopping malls, utilities, municipalities, businesses, and other entities [26].

In regard to the preferences of EV users with respect to the location of charging stations, it has been indicated that 65% of such users would prefer to have charging stations situated in residential areas, 41% in shopping malls, 39% in parking lots, and 38% in proximity to metro stations [22].

The latest research has concentrated on the subject of public charging. Nevertheless, it is observed that there is typically no comprehensive methodology for examining the charging patterns of EV users, including those utilizing public and residential charging facilities.

The study by Flammini *et al.* analysed data collected at 1750 public charging stations in th-e Netherlands over a 4-year period, while the study by Gellrich *et al.* [29] was based on public charging stations in Switzerland. Both studies examined the time-varying intensity of the use of public charging devices depending on location, which is posited that the demand will increase even further, particularly in light of the growing prevalence of EVs. Nevertheless, there is no discernible strategy for the utilization of rapid charging stations [30, 31].

Yang *et al.* employed a modelling approach to investigate the demand for fast charging (DC) based on a comprehensive dataset comprising over 2000 charging transactions collected over a 7-month period from 130 EVs in Beijing, China. In this context, it was posited that driving behaviours, weather conditions, and previous user habits with regard to conventional vehicles were identified as the principal factors affecting the charging behaviour of EV users [32].

In a recent study, Lee *et al.* examined the socio-economic factors influencing the charging behaviour of EV users in California. It is therefore asserted that the characteristics of the EV, the commuting behaviour of the user, and the availability of charging facilities at the workplace are significant factors influencing the user's choice of charging location [33].

In the study on charging behaviour in the literature on charging stations in Canada and the charging needs of EV users, a multidimensional comparison was conducted based on the analysis of fast and slow charging in public spaces and slow-charging concepts in homes. It can be observed that the average energy consumption at charging stations for public slow charging is 8.83 kWh, with an average charging time of approximately 3 hours and an average parking time of 5.15 hours. It has been established that the mean energy consumption is in excess of 12 kWh, while the mean charging time is approximately 10.5 hours when charging with private charging stations situated in residential buildings [34]. Furthermore, the data obtained from the fast-charging stations situated in public areas demonstrate that the mean energy consumption is 12.89 kWh, the average EV commences charging with a state of charge (SoC) level of 34%, and the charging process concludes with approximately 73%.

Furthermore, the study indicates that the commencement of charging at charging stations situated in public areas occurs predominantly between 7:30 a.m. and 8:30 a.m., with a secondary peak observed between 12:00 p.m. and 13:30 p.m. The commencement of charging at private charging stations in residential buildings typically occurs between 5 p.m. and 6 p.m. In contrast, charging at DC charging stations in public spaces is predominantly observed between 12:00 a.m. and 12:30 a.m., as well as between 4 p.m. and 5 p.m. [34].

Furthermore, it is highlighted that during periods of high parking at AC charging stations situated in public areas, which typically exceed 5 hours, EVs remain connected to the charging stations even when they are fully charged. This can result in other EV users being unable to utilize the charging stations or actively charging their vehicles, which ultimately reduces the overall service level and capacity utilization rates of the charging stations [35].

Caperello *et al.* identify the extended parking periods for EVs at charging stations, despite the vehicles' full battery capacity, as a novel manifestation of improper conduct in this context. He concluded that this acts as a deterrent to a greater number of EV users from utilizing public charging stations. It was also noted that the installation of charging stations can have a detrimental impact on the operational profitability and user comfort, as well as result in lengthy charging and parking times [36].

Furthermore, it has been demonstrated that there are discrepancies in the utilization of charging stations in residential edifices, particularly with regard to the seasonal charging necessities of EV operators. It has been observed that the number of charges, average energy consumption, and average charger connectedness parameters exhibited a notable increase in November, while there was a discernible decline in usage during the subsequent period, spanning December to January. In the locations where this study was conducted, it is stated that the relevant time periods coincide with the winter semester holiday period, which is associated with a reduction in transportation demand during this period.

In the context of the prevailing market conditions, the utilization of AC charging stations represents a more cost-effective alternative to the deployment of DC charging stations in the provision of charging services for EVs. Consequently, those who utilize EVs tend to favour AC charging stations when the requisite conditions are met. A simulation based on the average length of stay in a shopping mall was conducted to ascertain the utilization rate of an AC charger with three sockets in the context of approximately 350–400 EVs per day operating for 12 hours a day. The results indicated that the average utilization rate was 94.8%. However, when the number of sockets was increased to five, the utilization rate decreased to 77.9% [37].

The provision of direct support for the development of EV charging infrastructure by public authorities is imperative for the widespread adoption of sustainable transportation. In the initial diffusion phase, a period in which market mechanisms frequently prove ineffective, state-led incentives have been shown to accelerate infrastructure investments, thereby enhancing user confidence and mitigating investment risks for the private sector. Such public intervention not only facilitates the energy transition but also contributes to reducing regional disparities and achieving environmental policy objectives.

In particular, there are good examples of the approach you mention in Turkiye, such as Uşarj, an EV charging operator under the General Directorate of Turkish State Railways (https://usarj.com.tr/), and Şarj Park, an EV charging operator under Istanbul Metropolitan Municipality (https://enerji.istanbul/faaliyetlerimiz/enerji-yonetimi/yenilenebilir-enerji/sarjpark-elektrikli-arac-sarj-sistemleri/). There are no systematic studies in the literature based on the analysis of time-dependent usage data of charging stations in special-use commercial areas such as shopping malls in Turkiye. This study has highlighted the need for new empirical analysis in this context. The analysis was based on indicators such as usage intensity, average charging time, capacity utilization, and concurrent usage rates. These indicators were measured using time series data from charging stations in a shopping mall. An evaluation of the performance level of the obtained parameters is performed, and the charging rates are then analysed. Demand-based configuration recommendations are then provided.

As is the case in many areas, the potential for public support and subsidies in the management of EV charging stations is likely to yield highly effective results. However, in a fundamental sense, particularly in areas such as shopping malls, where sustainable and competitive conditions are quite intense, public operators in Turkiye are in place. The majority of public EV charging stations are located within the car parks of General Directorate of Turkish State Railways, in addition to those owned by the municipality in İstanbul Energy. A study on EV charging stations in public car parks may develop new concepts that can provide added value by addressing this important approach.

3 Evaluating charging station performance using real-world data

The widespread adoption of EVs is contingent upon the accessibility and affordability of charging infrastructure. This represents a substantial shift in sustainable transportation strategies, resulting in a swift surge in demand for charging infrastructure. In this context, public charging stations are strategically located in urban areas with the greatest potential for usage.

The proposed solutions that are discussed in this study focus on making the charging infrastructure for EVs more efficient and more sustainable. In particular, smart charging systems are characterized by advanced control and communication technologies that optimize the timing and power level of charging operations within the framework of the instantaneous state of the grid, the energy supply/demand balance, and user preferences. The proposed solutions discussed in this study focus on making the EV charging infrastructure more efficient and sustainable. As well as spreading energy consumption over time, these systems play a vital role in integrating renewable energy sources and balancing the load on the grid. Flexible pricing approaches, such as dynamic pricing models, are an important tool in this respect, as they not only spread energy consumption over time, but also play a crucial role in integrating renewable energy sources and balancing the load on the grid. These models involve flexible pricing strategies where the unit price of energy varies according to time, demand intensity, or market conditions,

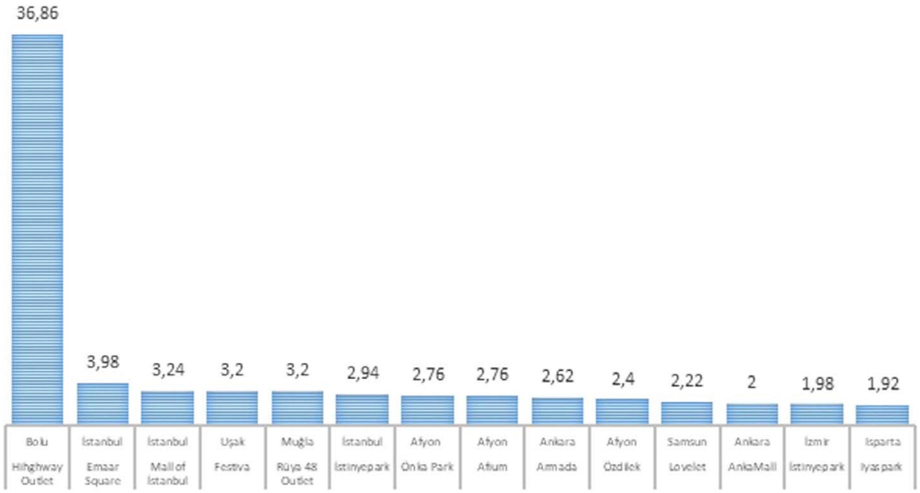

Figure 2. Turkiye's shopping malls with the highest DC socket output power (MW).

and they provide both economic benefits and support for grid stability by managing users' consumption patterns. These systems not only spread energy consumption over time, but also play a crucial role in integrating renewable energy sources and ensuring grid load balance.

In the context of the EV charging market in Turkiye, there is a growing competitive landscape, with an increasing number of actors seeking to expand the availability of public charging stations in order to meet the anticipated future demand. In contrast, the financial profitability of charging network operators has not reached the anticipated levels under the prevailing circumstances. These challenges include the high costs of establishing new charging stations, the underutilization of existing infrastructure, and the potential offered by emerging technologies in the market. Those currently operating charging networks are developing strategies with the objective of increasing customer loyalty and gaining a competitive advantage. The implementation of innovative business models and the pursuit of economies of scale through strategic location decisions will facilitate the enhancement of financial profitability.

It is of paramount importance to optimize the capacity utilization rates of charging stations situated in public spaces, such as shopping malls, which experience high levels of human and vehicular traffic. It is determined that the appropriate orientation of the relevant infrastructure investments and the efficient fulfilment of user requirements are contingent upon the effective utilization of charging stations. It is notable that the existing literature on the utilization rates of charging stations in shopping malls is limited. Furthermore, there is a dearth of open-source data that could inform more effective management of existing resources.

A total of 1403 DC charging sockets, with a cumulative installed power capacity of 302.05 MW, have been commissioned at various shopping mall locations across Turkiye, forming a significant component of the nation's high-power EV charging infrastructure. These installations account for approximately 11.3% of all DC charging sockets and 16.9% of the total installed DC charging capacity across Turkey, highlighting their disproportionately high contribution to the national high-power charging infrastructure [38].

In this respect, the DC charging stations in Turkiye were examined, with the objective of identifying the centres with the highest output power. This was determined on the basis of the shopping malls included in the study and their respective locations. The resulting data are presented in Fig. 2.

The present study presents the results of an empirical investigation conducted on a DC charging station with a 180-kW dual-plug output in a shopping mall situated in Isparta, Turkiye. The data analysis encompasses the temporal span of 30 September 2024 to 1 July 2023. A comprehensive analysis of the actual usage data corresponding to the relevant dates was conducted to examine in depth the capacity utilization rate of the EV charging station. The calculations were based on the results of the analysis and considered a number of factors, including the daily frequency of use of the relevant charging point, the average charging time, fluctuations in demand, and the charging preferences of EV users. In consideration of these parameters, it is our objective to present quantifiable data regarding the efficiency and performance of the current charging infrastructure. Furthermore, we will present strategies that are both feasible and applicable with respect to capacity expansion and operational enhancements.

In light of the findings yielded by the reference sample, our objective is to devise strategic solutions that will enhance the efficiency of charging infrastructures in shopping malls. The findings indicate that, beyond the analysis of the capacity utilization rate of charging stations in shopping malls, there is potential for the development of adaptable and flexible solutions that align with user demand. In light of the data obtained, concrete outputs were generated regarding the enhancement of operational efficiency with a view to optimizing the EV charging infrastructure. The results obtained serve as a basis for future charging station infrastructure investments.

In consequence, the findings of the investigation conducted in alignment with the data collected regarding the EV fast-charging facilities in shopping complexes are delineated below.

4 Unveiling insights on capacity utilization rates through analysis of real-world operating data

This study analyses the capacity utilization rate of the 180-kW dual-output DC charging station situated within a sample

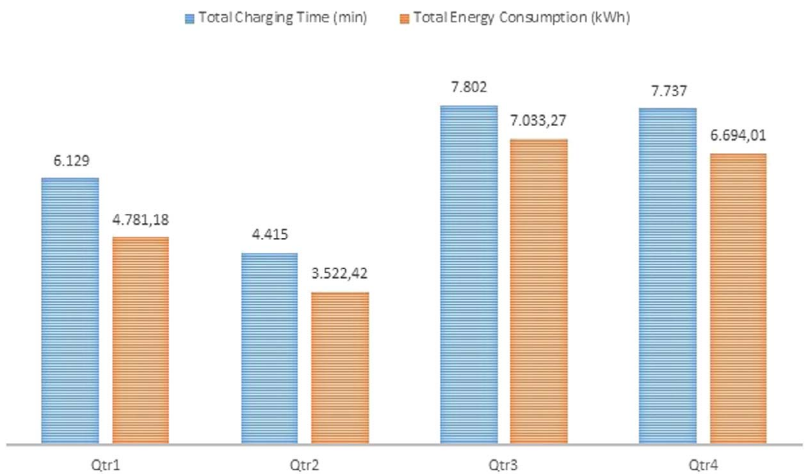

Figure 3. Charging rates and energy consumption of EVs by quarters of the year.

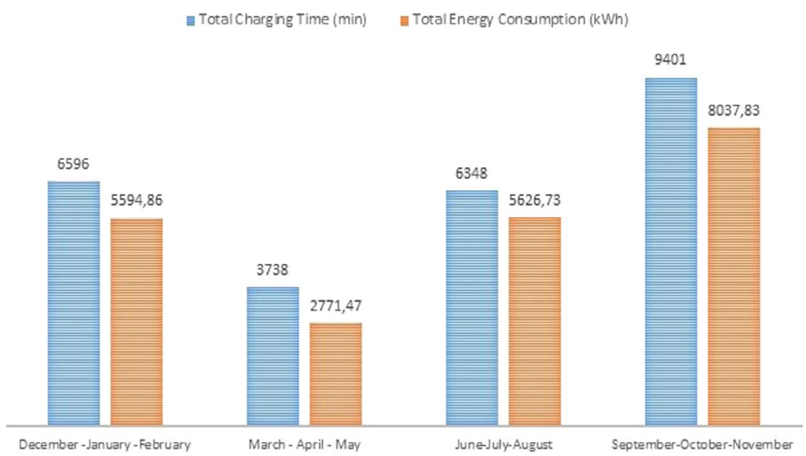

Figure 4. Seasonal trends in charging rates and energy consumption of EVs.

shopping mall. The objective of the analysis was to identify the principal factors influencing the utilization rates of the charging station. The fluctuations in user demand over the specified time period and the underlying factors influencing these fluctuations have been subjected to exhaustive analysis. In particular, it was found that the increase in the number of users resulted in an increase in the usage capacity of the charging station and revenue generated, particularly during periods of high usage. Conversely, the capacity utilization rate decreased significantly to zero during periods of closure. The findings offer charging station operators the chance to devise strategic plans to enhance their capacity utilization rates. These strategies have the potential to respond effectively to current user demands and to manage future infrastructure investments in a more efficient manner.

The initial analysis indicates that the utilization rates and energy consumption of EV charging stations exhibit seasonal fluctuations and are influenced by user behaviour.

Figure 3 illustrates a notable decline in the utilization of EV charging stations during the initial two quarters of the year, which is indicative of a corresponding reduction in energy consumption. In the final two quarters, the utilization time and energy consumption values of EV charging stations have exhibited a high degree of correlation.

Furthermore, the impact of seasonal variations on the utilization rates of EV charging stations was also examined. Low temperatures during the winter months have a detrimental impact on battery efficiency, resulting in increased energy consumption by vehicles. However, a reduction in travel due to inclement weather and low temperatures may also result in a decline in demand for charging stations. The relatively clement weather conditions that prevail during the spring and summer months result in a notable increase in travel, which in turn gives rise to a corresponding rise in the utilization of charging stations.

Figure 4 illustrates the seasonal variation in the utilization and energy consumption of a representative charging station. The data indicate that the highest rates of both charging station use and energy consumption occur during the autumn period. In accordance with conventional wisdom, the autumn period is viewed as a transitional phase in vehicle usage, particularly in light of the conclusion of intercity travel and holiday periods. However, in the dataset that forms the basis for this study, which examines a sample of charging stations situated within shopping malls, it is postulated that the pertinent period—which coincides with the commencement of the academic year—is informed by the proclivities of EV users in relation to their preparatory activities. In consequence, the number of visits to shopping malls has risen in accordance with the increasing demand. During the winter months, when charging stations are utilized with increased frequency, thereby resulting in elevated energy consumption,

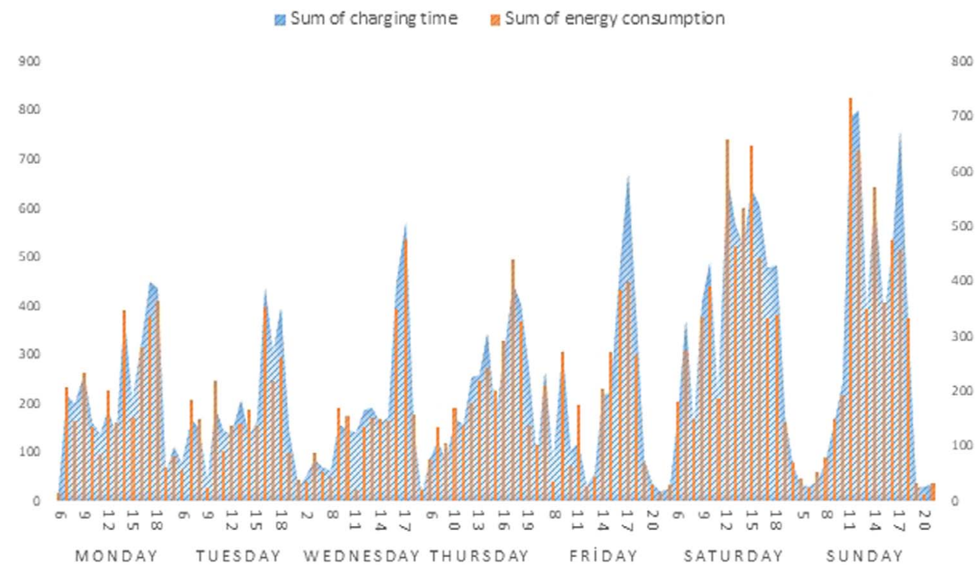

Figure 5. Daily average hourly usage trends of the charging station.

it is postulated that the frigid climatic conditions prevalent in Isparta exert considerable influence upon the proclivity of those who utilize EVs to visit shopping malls. Furthermore, during the spring season, when the lowest usage of the charging station (and consequently reduced energy consumption) was observed, it was concluded that users spent more time outdoors and were less likely to visit shopping malls. It was observed that, during the summer season, usage rates and energy consumption levels were similar to those recorded in the winter season.

The utilization of EV charging stations situated within shopping malls is contingent upon visitor density and user habits, with daily and hourly usage patterns exhibiting a direct correlation. The results of the analysis indicate that there is a generally low rate of usage observed on weekday mornings, which rises significantly with the increase in visits to shopping malls, particularly from the midday hours onwards. The highest demand for EV charging is typically observed during the period between 4 p.m. and 7 p.m., which coincides with the postwork period. However, on weekends, the usage rates of charging stations tend to remain consistently high throughout the day. The observed decline in the utilization of charging stations at shopping malls during the early morning period and in close proximity to closing time offers potential for optimizing capacity management and user demand. In periods of low demand, the utilization rates of charging stations could be increased by implementing incentive strategies for EV users. Such strategies might include discounted pricing or promotional campaigns. In this context, EV charging stations present a significant potential for utilization during 10 p.m. to 10 a.m., when shopping malls are closed. In particular, during a period when energy unit prices are at their lowest (in accordance with the multitariff system), the additional advantages afforded have the potential to facilitate further opportunities for both charging station operators and EV users. The comprehensive analysis conducted in this study facilitates the more efficient utilization of charging infrastructure and the effective management of energy resources. This is of particular importance for the enhancement of the profitability of charging station operators and the satisfaction of EV users.

Figure 5 below presents the hourly usage data of the charging station, reduced to a daily basis, covering the period from 1 July 2023 to 30 September 2024. The data presented here are based on the hourly time intervals commencing at the start of the charging process.

The analysis of the sample revealed that the charging station becomes active at 6:00 a.m. on weekdays, with the exception of Wednesdays. Based on the analysis of the findings, our investigation concludes that the designated charging station remains accessible during nonoperational hours of the respective shopping mall. This advantageous feature presents a significant operational benefit, distinguishing itself from other shopping malls by facilitating enhanced utilization capacity. Analysis of the acquired data indicates that the charging station's usage patterns exhibit a consistent termination of operations post-19:00 hours across the general weekly period. Analysis of daily utilization patterns reveals that while the charging station experiences increased activity during afternoon hours throughout the week, Mondays demonstrate the lowest operational efficiency. Conversely, when evaluating daily capacity utilization metrics, Sundays exhibit the highest utilization rates. It is notable that there is a considerable increase in the utilization of charging stations during the postmeridian period. Furthermore, there is a discernible correlation between this heightened usage and the elevated capacity ratios observed. Similarly, the charging stations exhibit notably high utilization rates on Sundays, with operational patterns indicating significant activity commencing at 9 a.m. and demonstrating a substantial decrease in usage following 5 p.m. The results demonstrate a considerable rise in both utilization intensity and capacity rates towards the end of the week, including Fridays, which is consistent with the anticipated patterns.

Figure 6 presents a comparative analysis of the average utilization rates of charging stations across different days of the week. A comparative analysis of charging station usage trends on a weekly day basis is presented. The graph reveals that the intensity increases on weekdays due to working days, while it shows a more balanced distribution on weekends. The occurrence of significant peaks according to usage hours during the day is considered to be directly related to the

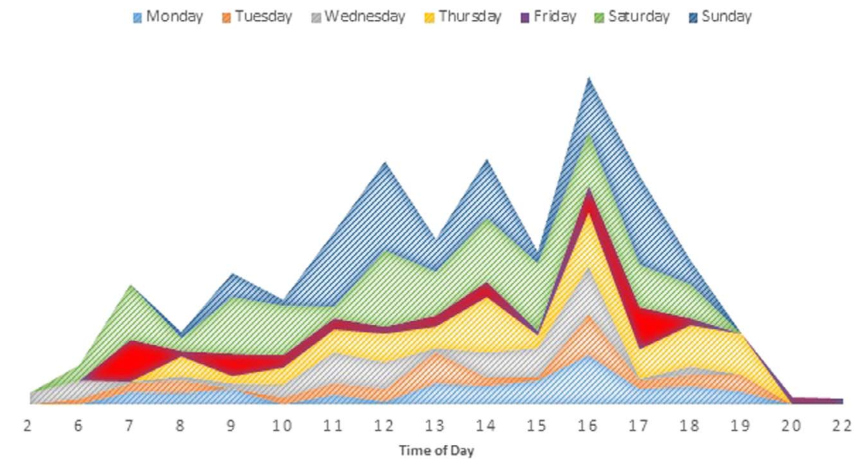

Figure 6. Comparative analysis of average daily usage trends of the charging station.

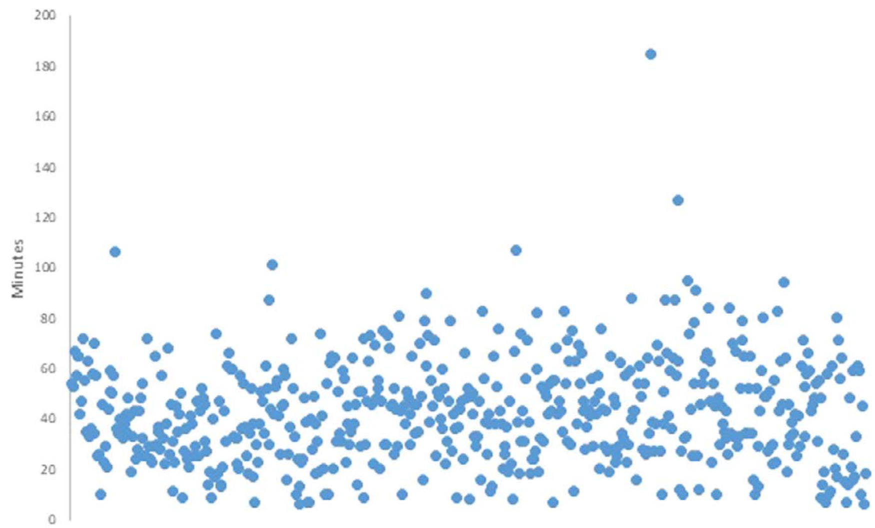

Figure 7. Duration of EV charging.

transport and daily routines of the station. The graph reveals the dynamic usage profile of charging stations, enabling a data-driven approach for future investments in EV infrastructure. The findings of this study provide critical data for the optimisation of EV infrastructure. This is especially evident in the domains of energy management and capacity planning, where the meticulous delineation of utilization patterns is poised to play a pivotal role in the formulation of sustainable urban policies.

Furthermore, an additional analysis was conducted to ascertain the mean charging times for EVs utilizing the charging station during the specified time frame. The mean charge time for EVs (exclusive of instances where the charge time was less than 5 minutes) was determined to be 42.46 minutes based on the data collected during the specified period. These results can be found in Fig. 7.

A review of the data indicated that the vehicles utilized an average of 33.68 kWh of energy for charging, with an average battery SoC reaching up to 52%. In consideration of the findings presented in Fig. 8, an evaluation has been conducted to ascertain the suitability of this charging station for use with a range of EVs from various manufacturers and models. The results of our investigation suggest that a subset of vehicles display a proclivity towards a relatively consistent battery capacity and charging patterns. Nevertheless, it was not feasible to determine the battery capacities based on the observed battery SoC levels.

The mean energy consumption per vehicle was calculated to be 35.78 kWh, with a total of 16.5 MWh consumed across 463 charging events over the entire period under consideration. The results revealed that the total energy consumed by each vehicle ranged between 86.75 and 2.82 kWh, with an overall mean value of approximately 19.5 kWh. It has been determined that the cost of charging is a fixed rate of €0.23/kWh (On 22 May 2025, the Turkish Central Bank set the exchange rate for the Euro at €1 = 43.86 TL.).

The data presented in Fig. 9 depict the energy consumption values per vehicle. This represents a significant potential, particularly in light of the current utilization of the charging station. The specifics of the innovative approaches will be discussed in greater detail at a later stage. The graph illustrates various vehicle types; however, certain data points, particularly those pertaining to battery capacity and type, are nonseparable. The acquisition of this data would undoubtedly yield a highly valuable set of results. The investigation could have yielded intriguing findings on the health status of the batteries if the potential for variation in charging times for vehicles of the same make and model had been considered.

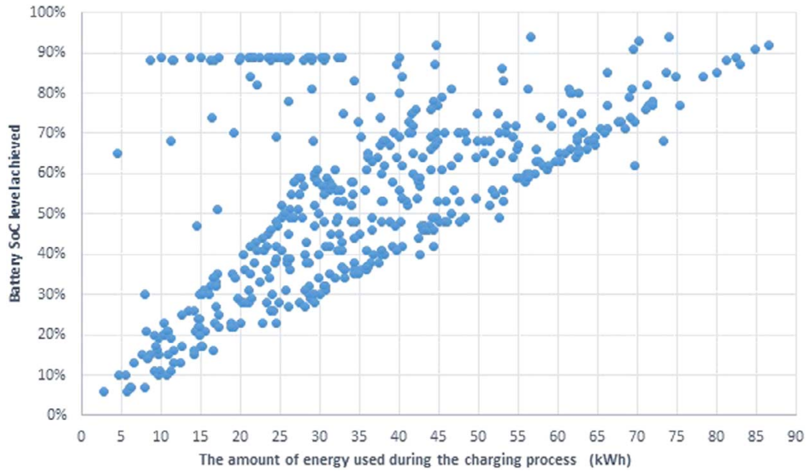

Figure 8. Battery capacity utilization and energy consumption.

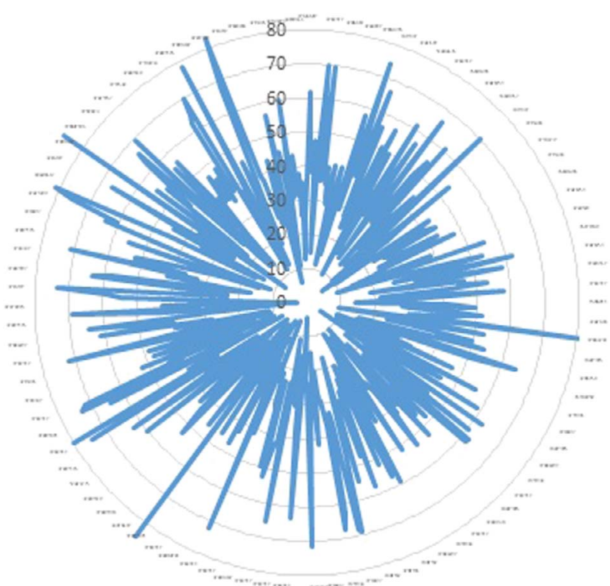

Figure 9. Analysis of energy consumption per vehicle unit.

However, it should be noted that the provision of such data is not technically feasible.

Also, Fig. 10 illustrates that for EV batteries with a maximum power capacity of 180 kW from the charging station, the average battery charging time is calculated to be 1.48 minutes/kWh. Given the variability in maximum charging power and charge/discharge curves across different EV system designs, it is not feasible to provide a battery capacity approach for EVs utilizing this charging station.

The charge start and charge end levels (SoC, %) for EVs utilizing this charging station are illustrated in Fig. 11. Consequently, the results of the analysis, based on the battery SoC levels, demonstrate that the commencement of charging is initiated at an average of 30% and concluded at an average of 75% for EVs utilized within the specified time interval, as evidenced by the sample.

The graph illustrates the utilization rates of the two plug outputs in the 180-kW DC charging station. The results presented in Fig. 12 indicate that the potential usability of the charging station can be evaluated based on the standard usage times of the shopping mall between 10 a.m. and 10 p.m. on a monthly basis. In light of the significance attributed to graphical results, it has been established that the aforementioned charging station may be utilized during periods of closure of the shopping mall, in accordance with standard operating procedures. It is therefore concluded that the general utilization potential can be increased. Nevertheless, it has been noted that the utilization of the charging station markedly declines during the periods when the shopping mall is not in operation.

The 180-kW dual-output DC charging station, upon which the sample in the study is based, provides a rapid charging solution that is currently the fastest in its field. If the system architecture of the EV in question is appropriate, it has the potential to significantly reduce the average charging time on a vehicle-by-vehicle basis. In this context, the time required for charging and the costs associated with charging are crucial parameters that directly impact the user experience, particularly in terms of usage efficiency. The analyses offer significant insights into the impact of charging duration and associated costs on station utilization rates.

Figure 13 illustrates the mean charging time and the mean energy consumption for on an hourly basis output sockets. The figure illustrates the data on the utilization of charging stations on a socket basis. In this study, the hours of the day were designated as the sampling period, and the average usage rate and energy consumption amounts were analysed. According to these data, it is observed that the usage time of the second sockets is relatively low, as evidenced by the previous graph. Furthermore, partial usage was identified in the time period until 10:00, when the shopping mall was closed.

5 A critical examination of the findings and an investigation into potential future implementation strategies

As the number of EVs on the road continues to grow, the varying charging times, durations, and locations of these vehicles present a challenge to the integration of the energy grid and the cost of energy per unit. The phenomenon of uncontrolled charging has the potential to result in an increase in the unit costs of the charging service, particularly when such activity occurs during periods of elevated electricity demand. This may result in the operation of charging stations with low capacity factors, providing charging services exclusively during peak hours and necessitating further investments in

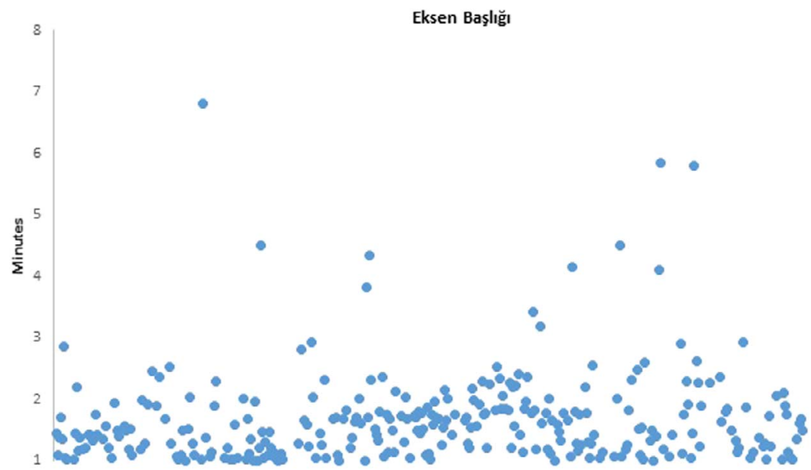

Figure 10. Charging duration comparison by vehicle type for 1 kWh at 180-kW DC station.

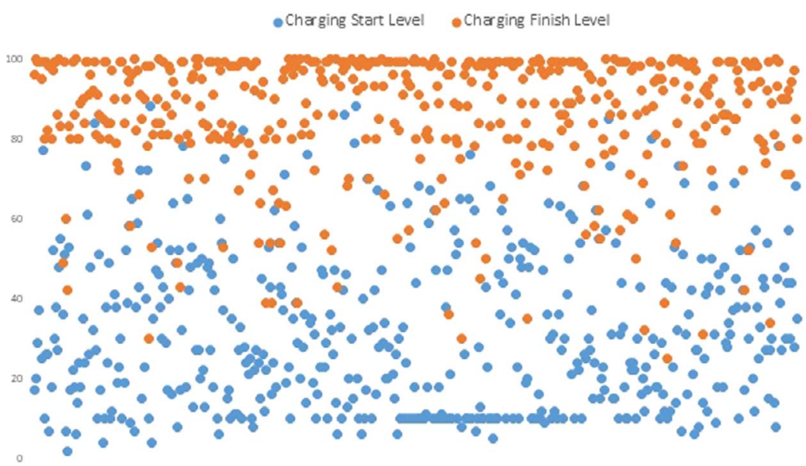

Figure 11. Charging activation and deactivation patterns (%) of EVs at charging stations.

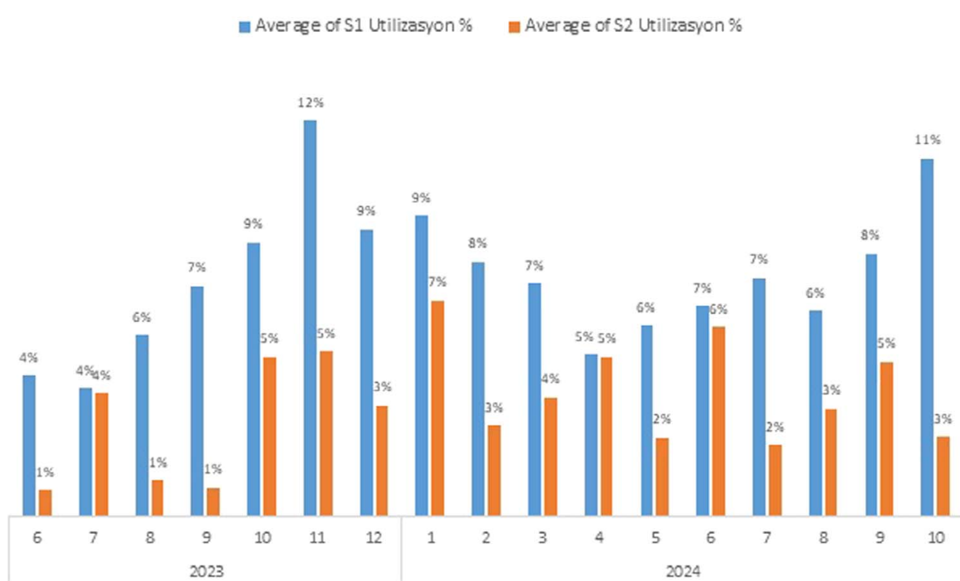

Figure 12. Usage rate of charging stations by sockets: monthly distribution analysis.

energy networks. It is therefore evident that the charging methods employed for EVs play a pivotal role in the management of energy costs and the development of infrastructure requirements.

The assessment, development, and implementation of novel business models for the operation of EV charging stations is inextricably linked with the advancement of EV charging infrastructure and other developments within EV ecosystems.

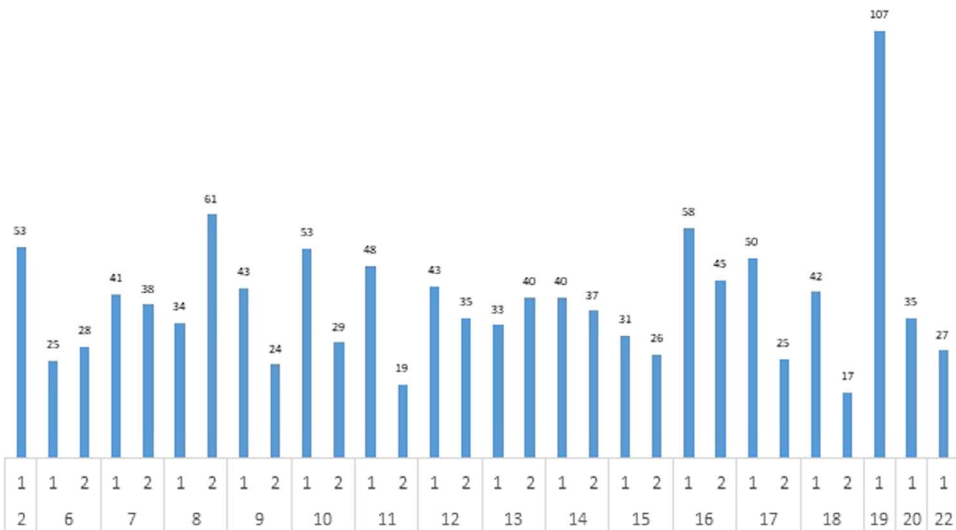

Figure 13. Average charging duration by socket type: a comparative analysis.

The sustainability of investments made and planned in this area represents a significant challenge for many businesses, particularly in light of the necessity for supplementary state-funded financial subsidy programmes.

In accordance with the EV and Charging Infrastructure Projection Report, as prepared by EMRA, the projection until 2035 under three basic scenarios (low, medium, high) has been published. In accordance with the aforementioned projections, the number of charging points for EVs is expected to reach 34 278–61 897 by 2025, 83 543–181 274 by 2030, and 146 916–347 934 by 2035 [6]. In the context of the evolving EV ecosystems in Turkiye, it is postulated that the implementation of high-efficiency approaches through the proposed model for charging infrastructure investments will enhance the sustainability of these activities.

The objective of this study is to raise awareness among operators of charging stations in Turkey of the charging requirements of EV users, based on the concept of smart charging. This paper is based on the application of dynamic pricing models for the purpose of charging service unit costs and the alignment of charging behaviour in accordance with retail electricity selling prices during the course of a day. In line with the objective of offering an indirect incentive to those who utilize EVs, the intention is to provide savings in the basic operating costs of EV users. This will be achieved through the implementation of optimal solution approaches during periods when the grid load is low or the electricity generated by renewable energy sources is high. This will be based on electricity pricing strategies.

The energy consumed on the grid side is contingent upon the orientation of the charging requirements for load management. Despite the highest penetration rates of the EV ecosystem in certain markets, there are still areas with low utilization rates of charging stations. Indeed, it has been demonstrated that even in those countries within the European Union where this ecosystem is the most advanced, the maximum capacity utilization factor of charging points in public spaces can reach 30% [39]. A study prepared by McKinsey indicates that in order to operate a charging network in an economically viable manner, a capacity utilization rate of approximately 15% is generally required [40, 41]. It is anticipated that the utilization rate will increase to 7.2% or 1.72 hours for slow chargers and 17.6% or 4.22 hours for fast chargers. In this context, it is evaluated that effective monitoring of the usage data of existing charging stations and making the data accessible are critical for the improvements to be made [42].

In alignment with the methodology delineated in this study, the following strategies are recommended for implementation by the pertinent stakeholders. The results of the study suggest that the current utilization pattern of the charging station infrastructure may not be the most cost-effective. In light of the aforementioned findings, a series of recommendations have been put forth with the aim of enhancing the efficiency of EV charging stations. The implementation of strategies such as dynamic pricing models and demand-adjustable charging times can facilitate the optimal utilization of available capacity. Furthermore, it is recommended that applications be developed to facilitate shorter charging times and flexible reservation systems, with the objective of enhancing the user experience.

- The relocation of EV charging activities to off-peak energy demand periods, accompanied by the implementation of time-varying energy unit costs that align with the actual costs associated with electricity generation and distribution, represents a potential solution; this results in gains and stability for both EV users and producers/ distributors on the electricity grid side, as well as in centralized load distribution.
- The objective is to develop and implement facilitation practices for end users in the electricity distribution system, with a particular focus on external electricity metre subscriptions. This is with a view to incentivizing EV users with individual charger facilities, with the ultimate goal of reducing the multitime electricity unit price tariffs. These tariffs are currently determined by EMRA and included in the electricity unit price tariffs.
- The proposal is to implement a pricing strategy for energy unit costs that considers temporal and spatial variables. This strategy would include the integration of charging stations into the existing system, particularly in public areas such as shopping malls, where the demand for charging is relatively low or nonexistent during certain periods, such as when the shopping mall is closed.

- The objective is to enhance the flexibility of the electric power grid by enabling the return of the energy stored in EV drive batteries to the grid during periods of peak demand. This will be achieved by utilizing charging stations that offer external energy storage in compliance with the V2G standard for EVs.
- The primary objective is to present a comprehensive concept that incorporates a distinctive pricing strategy for regionally based charging service expenses.

Furthermore, EV charging stations are regarded as a potential source of overloading of lines and transformers, which may result in voltage imbalances, particularly in electricity distribution networks. It is possible that the increased use of electrical vehicles (with the consequent necessity for more extensive recharging facilities) may lead to certain restrictions being placed on a regional level. The issue of overloads (in the context of charging stations located in shopping malls, business centres, etc.) can be addressed through the implementation of regional pricing strategies and routing to feeders with lower energy demand. In electricity distribution systems, where there is a need to enhance network flexibility and reliability, it is recommended that the current technical constraints on the energy supply and demand sides be reduced and that costs be accurately reflected in energy unit prices. Additionally, the cost savings associated with the charging service provided by EV charging stations situated or to be situated in areas where grid energy demand is relatively low should be considered as a foundation for the model presented in this study. The proposed model is expected to enhance the efficiency and sustainability of EV charging infrastructures by integrating the aforementioned strategies.

6 Concluding synthesis critical findings implications and future perspectives

It is of paramount importance to gain an in-depth understanding of the charging behaviour of EV users in order to accurately predict energy demand and to guarantee comfort when using EVs. It is imperative that the planning of studies for EV charging infrastructure should not only result in increased profitability but also contribute towards the optimal stability of the electricity grid. In this regard, it is essential to adopt a multivariate and holistic approach to comprehend the underlying factors influencing charging behaviours. The proposed approach will result in more efficient future infrastructure investments, offering advantages in terms of both energy management and sustainability.

The evaluation, development, and implementation of new business models for the charging requirements of EVs is contingent upon developments in the EV charging infrastructure and ecosystems in which the EV market is expanding. The establishment of a business model that ensures the sustainability of these investments represents a significant challenge for many charge operating companies, thereby necessitating the implementation of supplementary state-sponsored financial subsidy programmes. The public sector plays a pivotal role in providing financial backing for investors in emerging EV ecosystems, particularly in the context of charging infrastructure. It is widely accepted that the public sector must provide support in order to establish a self-sufficient market. However, it is anticipated that this will not be a sustainable strategy in the long term.

The methodology presented in this study is founded upon the implementation of cost-effective strategies for the positioning and planning of charging infrastructure, with the objective of enhancing the capacity factor of existing investments in EV charging infrastructure.

In the context of electricity pricing strategies, it is proposed that slow- and fast-charging stations be installed in currently unused shopping malls between 10 p.m. and 6 a.m. This timeframe corresponds to the night peak in terms of grid utilization. This would enhance energy efficiency during periods when the grid is underutilized and optimize the requirements for EV charging during periods when the cost of energy is lower.

During the diurnal period, the system exhibits a reduced capacity for accommodating temporal variations in energy demand. Conversely, it demonstrates enhanced potential for integrating electrical energy generated from sustainable sources, particularly solar energy. However, the cost of energy produced by this source is typically higher compared to the peak energy demand observed during nocturnal hours. It is anticipated that the implementation of fast- and slow-charging stations in shopping malls will facilitate the effective utilization of these facilities.

The primary rationale for the implementation of these charging stations is to provide an alternative charging solution for EV users during periods of low demand at shopping malls. These periods are typically between 10:01 p.m. and 5:59 a.m., as well as between 6:01 a.m. and 4:59 p.m., based on the respective night and day schedules. It is therefore evident that, in periods of low grid load or high renewable energy generation, it would be prudent to incentivise cost-effective charging processes that do not increase grid load, as this would align with the prevailing market conditions. It is anticipated that the provision of incentives for the charging of EVs during periods of low energy demand will contribute to the stability of the grid as a whole. The pricing of these incentives will be determined by the actual costs of generation and distribution of electricity.

This study represents the inaugural comprehensive examination of the charging processes associated with EVs in Turkiye. The data gathered from EV charging stations was utilized to facilitate this investigation. The relationship between the total charging time, the time required for the battery to remain connected to the charger, and the time for the battery to reach the targeted level is noteworthy. A noteworthy relationship can be observed between several variables pertaining to the charging of EVs, including the total charging time, the time spent maintaining a connection with a charging point, the time required for the battery to reach the desired level of charge, the utilization rate of the charging points at a given station, the energy consumption, and finally, the utilization rate of the charging stations themselves in the context of fast-charging. It is recommended that future research examine these parameters in greater detail and develop alternative optimisation strategies to enhance the efficiency of fast-charging infrastructures in shopping malls. Moreover, further research could be conducted into financial business models based on dynamic pricing strategies. It is proposed that these models have the capacity to implement algorithmic (software-based) interactions between charging stations. One such interaction could involve the regulation of the unit price at one station when it is full, while another with the same capacity remains available. Alternatively, the promotion of competition could be implemented when there are several stations with the same capacity. It is proposed that new business models and perspectives be introduced to facilitate the intensified use of EVs and support sustainable charging management.

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Author contributions

Orhan Topal (Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing—original draft, Writing—review & editing)

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